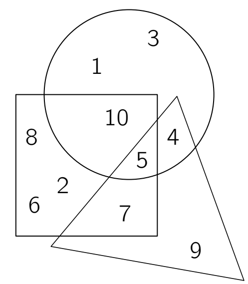
CANADIAN MATH KANGAROO CONTEST PROBLEMS

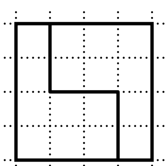
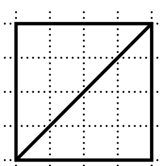
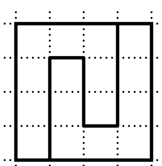
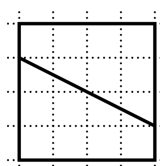
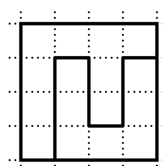
PART A: EACH CORRECT ANSWER IS WORTH 3 POINTS

1. Which number is inside the triangle, the square and the circle?

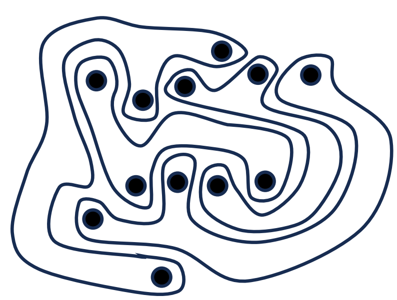


(A) 1 **(B)** 4
(C) 5 **(D)** 9
(E) 12

2. Which square is cut into 2 different shapes?

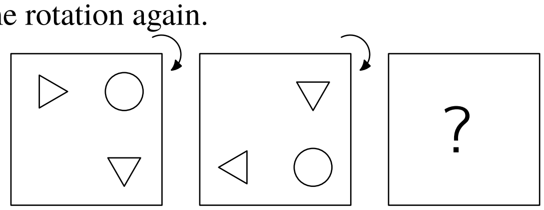
(A)  **(B)**  **(C)**  **(D)**  **(E)** 

3. The picture shows 4 strange shapes.
How many shapes have 3 dots inside?

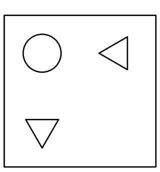
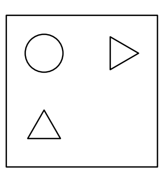
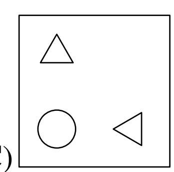
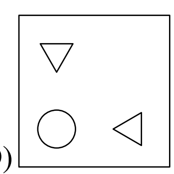
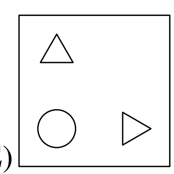


(A) 0 **(B)** 1
(C) 2 **(D)** 3
(E) 4

4. Kevin the kangaroo puts a picture on the table. He rotates the picture through a quarter turn, as shown. He then does the same rotation again.



What does Kevin see now?

(A)  **(B)**  **(C)**  **(D)**  **(E)** 

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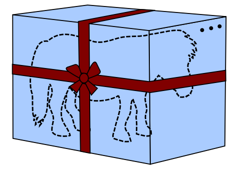
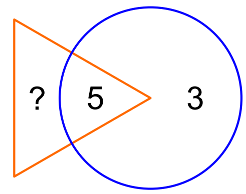
5. A toy pony is inside a box that is 1 metre tall, 1 metre wide and 2 metres long. A ribbon goes around the box, as shown. The knot uses an extra 1 metre of ribbon.
How long is the ribbon in total?



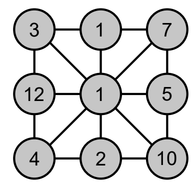
(**A**) 9 metres (**B**) 11 metres (**C**) 13 metres (**D**) 15 metres (**E**) 17 metres

6. The sum of the numbers in the triangle should be twice the sum of the numbers in the circle.
What number must replace the question mark?



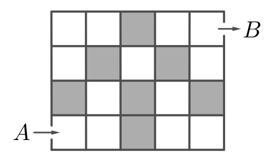
(**A**) 3 (**B**) 5 (**C**) 8 (**D**) 11 (**E**) 16

7. One of the numbers in the picture is equal to the sum of the numbers connected directly to it.
Which number is this?



(**A**) 3 (**B**) 5 (**C**) 7 (**D**) 10 (**E**) 12

8. Zara wants to move through the grid from A to B. She can only move to the right or upwards. Each time she visits a grey box, she has to pay 1 euro. Each time she visits a white box, she has to pay 2 euros.
How much would she pay for the cheapest path?

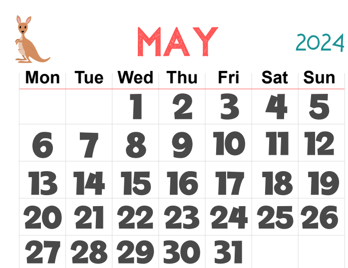


(**A**) 11 euros (**B**) 12 euros (**C**) 13 euros (**D**) 15 euros (**E**) 16 euros

9. A three-digit palindrome is a number of the form $'aba'$ where the digits a and b can either be the same or different. What is the sum of the digits of the largest three-digit palindrome that is also a multiple of 6?

(**A**) 16 (**B**) 18 (**C**) 20 (**D**) 21 (**E**) 24

10. Julia has a list of problems to finish during May.
She starts on the 1st of May. If she solves exactly 2 problems each day, she will finish the task on a Sunday.
If she solves exactly 3 problems each day, she will finish the task on a Wednesday.
How many problems are on the list?



(**A**) 6 (**B**) 12 (**C**) 18 (**D**) 24 (**E**) 30

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PART B: EACH CORRECT ANSWER IS WORTH 4 POINTS

11. What is the value of $\dfrac{20 \times 24}{2 \times 0 + 2 \times 4}$?

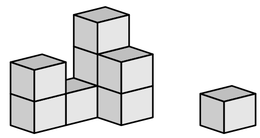
(**A**) 12 (**B**) 30 (**C**) 48 (**D**) 60 (**E**) 120

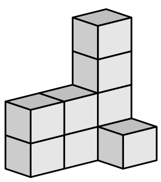
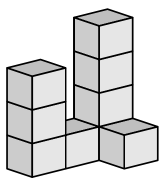
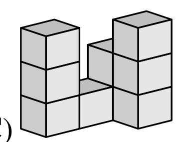
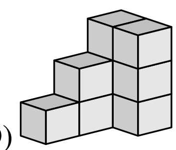
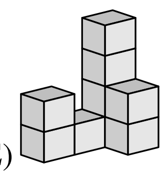
12. Ali, Bella, Che and Dimitry each have 3 shapes.
Each child has exactly one shape in common with every other child.
Which shapes does Dimitry have?

Ali	△	○	□
Bella	♡	□	☆
Che	☆	△	◇

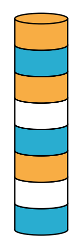
(**A**) □ ♡ ◇ (**B**) ♡ ○ △ (**C**) ☆ ◇ ○ (**D**) ◇ ○ ♡ (**E**) □ ☆ △

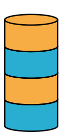
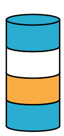
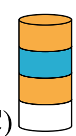
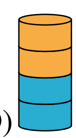
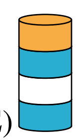
13. A cat knocks off a block from Felix's construction.
What could this construction have looked like **before** the block was knocked off?



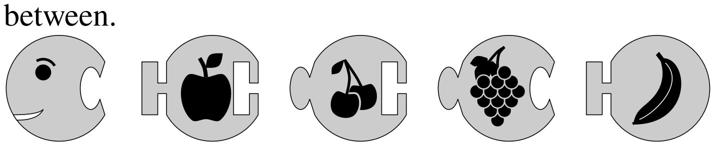
(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

14. Ada has built a tower of 8 discs, as in the figure. Ada removes the second disc from the bottom of this tower. Then she removes the third disc from the bottom of the new tower. Then she removes the fourth disc from the bottom of the new tower. Then she removes the fifth disc from the bottom of the new tower.
Which tower does Ada end up with?



(**A**)  (**B**)  (**C**)  (**D**)  (**E**) 

15. Lucas wants to make a caterpillar that has a head, a tail and either 1, 2 or 3 puzzle pieces in between.



How many different caterpillars can Lucas make without flipping pieces?

(**A**) 3 (**B**) 4 (**C**) 5 (**D**) 6 (**E**) 7

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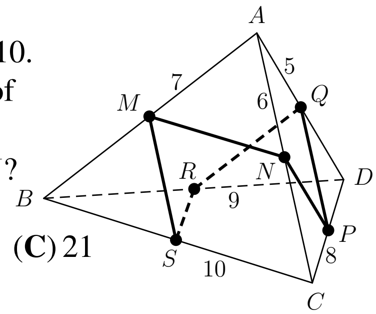
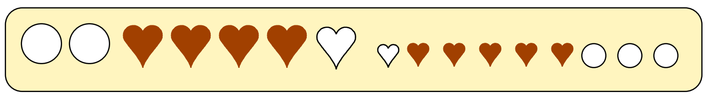
16. A triangular pyramid $ABCD$ has sides of length 5, 6, 7, 8, 9 and 10. The points M, N, P, Q, R and S are the midpoints of the edges of the pyramid, as shown.
What is the perimeter of the closed hexagonal line $MNPQRSM$?



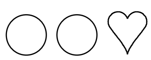
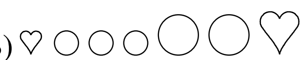
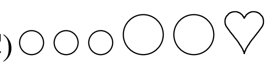
(**A**) 19 (**B**) 20 (**C**) 21 (**D**) 22 (**E**) 23

17. Three girls go to the tray one after the other and take some cookies.

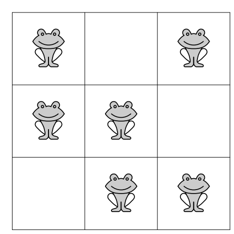


One of the girls takes all the hearts available on the tray. Another girl takes all the white cookies available on the tray. Another girl takes all the large cookies available on the tray. However, they do not necessarily take the cookies in this order. One girl takes three cookies, one takes six cookies and one takes seven cookies.

Which of the following sets of cookies does one of these girls take?

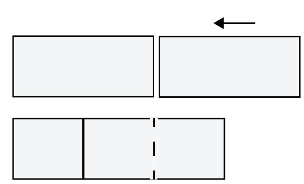
(**A**)  (**B**)  (**C**) 
(**D**)  (**E**) 

18. There are exactly 2 frogs in each row and each column. The frogs decide that 2 of them will jump to a neighbouring empty cell at the same time. Neighbouring cells have a side in common. After that, there still are exactly 2 frogs in each row and in each column.
In how many ways can the frogs do this?



(**A**) 1 (**B**) 2 (**C**) 3 (**D**) 4 (**E**) 5

19. Two identical rectangles, each with an area of 18, overlap to form a new rectangle, as shown. The new rectangle can be divided into three identical squares.
What is the area of the new rectangle?



(**A**) 24 (**B**) 27 (**C**) 30 (**D**) 32 (**E**) 36

20. Cristina has a set of cards numbered 1 to 12. She places eight of them at the vertices of an octagon so that the sum of every pair of numbers at opposite ends of an edge of the octagon is a multiple of 3. Which numbers did Cristina not place?

(**A**) 1, 5 , 9, 12 (**B**) 3, 5, 7, 9 (**C**) 1, 2, 11, 12 (**D**) 5, 6, 7, 8 (**E**) 3, 6, 9, 12

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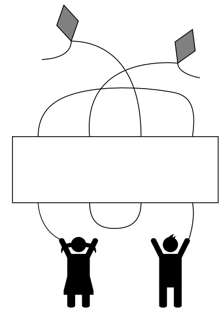
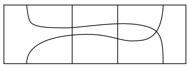
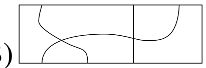
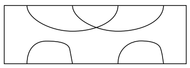
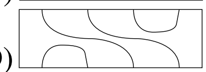
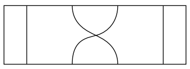
PART C: EACH CORRECT ANSWER IS WORTH 5 POINTS

21. Which of the strips should be placed in the space in the picture so that each child is connected to a different kite?



(A)  **(B)** 

(C)  **(D)** 

(E) 

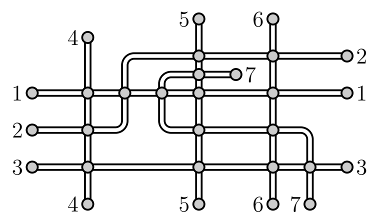
22. The Braille system for blind people, when written down, has the digits 0 to 9 represented by a set of black or white dots, as shown.

○● ●● ○○ 0	●○ ○○ ○○ 1	●○ ●○ ○○ 2	●● ○○ ○○ 3	●● ○● ○○ 4
●○ ○● ○○ 5	●● ●○ ○○ 6	●● ●● ○○ 7	●○ ●● ○○ 8	○● ●○ ○○ 9

How many different two-digit numbers contain exactly five black dots?

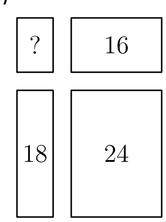
(A) 16 **(B)** 18 **(C)** 30 **(D)** 32 **(E)** 34

23. The figure shows the plan of the seven train routes of a small town. The circles indicate the stations. Martin wants to paint the lines in such a way that if two lines share a common station, then they are painted with different colours.
What is the smallest number of colours that he can use?



(A) 3 **(B)** 4 **(C)** 5 **(D)** 6 **(E)** 7

24. Gerard cuts a large rectangle into four smaller rectangles. The perimeters of three of these smaller rectangles are 16, 18 and 24, as shown in the diagram.
What is the perimeter of the fourth small rectangle?



(A) 8 **(B)** 10 **(C)** 12 **(D)** 14 **(E)** 16

25. Olya walked in the park. She walked half of the total time at a speed of 2 km/h. She walked half of the total distance at a speed of 3 km/h. She walked the rest of the time at a speed of 4 km/h. For what fraction of the total time did she walk at a speed of 4 km/h?

(A) $\frac{1}{14}$ **(B)** $\frac{1}{12}$ **(C)** $\frac{1}{7}$ **(D)** $\frac{1}{5}$ **(E)** $\frac{1}{4}$

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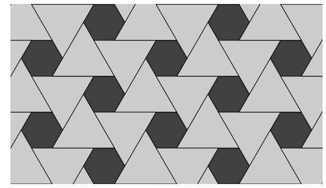
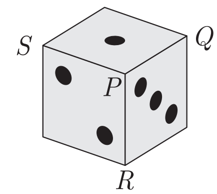
26. Teri the tiler is planning to make a large, square mosaic floor with a repeating pattern, using hexagonal and triangular tiles, arranged as shown in the diagram. She thinks she will use 3000 hexagonal tiles to make the whole floor.
Approximately, how many triangular tiles will she need?



(**A**) 1000 (**B**) 1500 (**C**) 3000 (**D**) 6000 (**E**) 9000

27. The number of the dots on opposite faces of a die add to 7. The vertex labelled P on the die is formed by the faces which have 1, 2 and 3 dots on them. Its vertex sum is the sum of the number of dots on those faces which meet at a given vertex. The vertex sum of P is $1 + 2 + 3 = 6$.
What is the maximum of the vertex sums of vertices Q, R and S?



(**A**) 7 (**B**) 9 (**C**) 10 (**D**) 11 (**E**) 15

28. There are n distinct lines on the plane, labeled $\ell_1, \ldots, \ell_n$. The line ℓ_1 intersects exactly 5 other lines, the line ℓ_2 intersects exactly 9 other lines, and the line ℓ_3 intersects exactly 11 other lines. Which of the following is a smallest possible value of n?

(**A**) 11 (**B**) 12 (**C**) 13 (**D**) 14 (**E**) 15

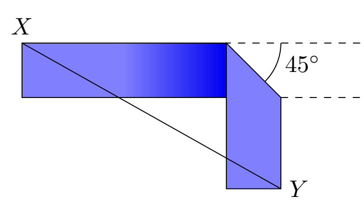
29. Kangaroo solves the equation $ax^2 + bx + c = 0$, and Beaver solves the equation $bx^2 + ax + c = 0$, where a, b, c are pairwise distinct non-zero integers. It turns out that the equations share a solution. Which of the following must be true?

(**A**) $a > 0$ (**B**) $b < 0$

(**C**) $a + b + c = 0$ (**D**) The common solution must be 0.

(**E**) The quadratic equation $ax^2 + bx + c = 0$ has exactly one real solution.

30. I have a strip of paper that is 12 cm long and 2 cm wide. I make a crease across it at 45° and then fold it, so that the two parts of the strip are aligned in a right angle, as shown.
What is the smallest possible length, in cm, of XY?



(**A**) $6\sqrt{2}$ (**B**) $7\sqrt{2}$ (**C**) 10 (**D**) 8 (**E**) $6 + \sqrt{2}$

Answer Keys: CEECBDCCED DDEBBCEDBE DCABADDBCB

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